Version 7.4.6

PVsyst - Simulation report

Grid-Connected System

Project: Gibraltar_Fixed_Solar_Panel

Variant: Gibraltar Report Simulation - Simulation for year no 25

No 3D scene defined, no shadings

System power: 14.99 MWp

Catalan Bay - Gibraltar

Author

University of Edinburgh (United kingdom)

Project summary

Geographical Site		**Situation**		**Project settings**	
Catalan Bay		Latitude	36.14 °N	Albedo	0.20
Gibraltar		Longitude	-5.34 °W		
		Altitude	176 m		
		Time zone	UTC+1		

Weather data
Catalan Bay
Meteonorm 8.1 (1996-2015), Sat=100% - Synthetic

System summary

Grid-Connected System
Simulation for year no 25

No 3D scene defined, no shadings

PV Field Orientation		**Near Shadings**	**User's needs**
Fixed plane		No Shadings	Monthly values
Tilt/Azimuth	32.2 / 0 °		

System information

PV Array		**Inverters**	
Nb. of modules	25852 units	Nb. of units	4 units
Pnom total	14.99 MWp	Pnom total	12.50 MWac
		Pnom ratio	1.200

Results summary

Produced Energy	19205923 kWh/year	Specific production	1281 kWh/kWp/year	Perf. Ratio PR	62.05 %
Used Energy	26989774 kWh/year			Solar Fraction SF	32.65 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation
Fixed plane
Tilt/Azimuth 32.2 / 0 °

Sheds configuration

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Average Height 15.3 °

Near Shadings
No Shadings

User's needs
Monthly values

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year	
3011	2748	2913	2721	2027	1745	1925	2061	1971	2159	2061	1648	26990	MWh

PV Array Characteristics

PV module
Manufacturer Generic
Model NB-JD580
(Original PVsyst database)
Unit Nom. Power 580 Wp
Number of PV modules 25852 units
Nominal (STC) 14.99 MWp
Modules 1124 string x 23 In series

At operating cond. (50°C)
Pmpp 13.88 MWp
U mpp 923 V
I mpp 15035 A

Inverter
Manufacturer Generic
Model SG3125-HV-20
(Original PVsyst database)
Unit Nom. Power 3125 kWac
Number of inverters 4 units
Total power 12500 kWac
Operating voltage 875-1300 V
Max. power (=>25°C) 3593 kWac
Pnom ratio (DC:AC) 1.20

Total PV power
Nominal (STC) 14994 kWp
Total 25852 modules
Module area 66782 m²
Cell area 61655 m²

Total inverter power
Total power 12500 kWac
Max. power 14372 kWac
Number of inverters 4 units
Pnom ratio 1.20

Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 1.0 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

LID - Light Induced Degradation
Loss Fraction 2.0 %

Module Quality Loss
Loss Fraction -1.3 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.2 %

Module average degradation
Year no 25
Loss factor 0.4 %/year

Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

System losses

Unavailability of the system

Time fraction 2.0 %
7.3 days,
3 periods

Horizon definition

Horizon from Meteonorm web service, lat=36.1354, lon=-5.3435

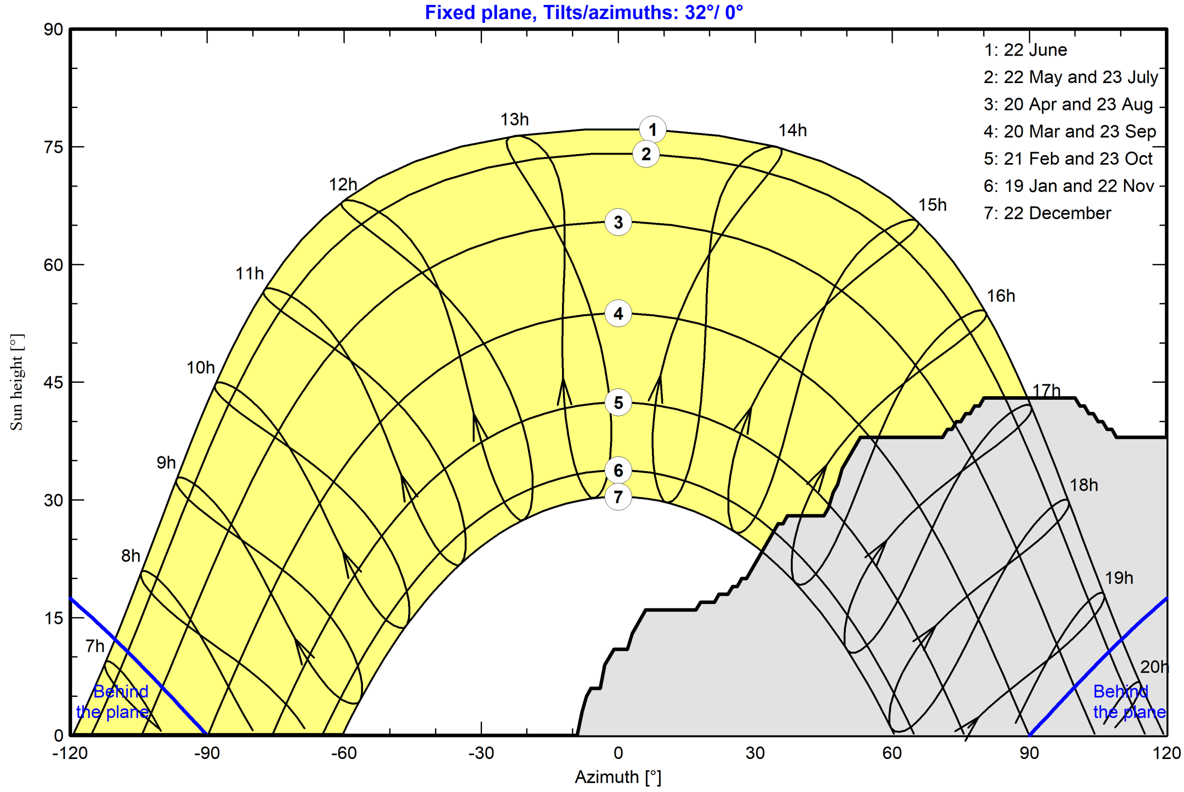
Average Height	15.3 °	Albedo Factor	0.53
Diffuse Factor	0.84	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-178	-174	-173	-167	-150	-149	-8	-7	-3	-1	2	6	17
Height [°]	12.0	12.0	10.0	8.0	1.0	1.0	0.0	3.0	5.0	9.0	11.0	11.0	16.0	16.0
Azimuth [°]	22	24	25	27	28	35	37	45	46	48	49	53	72	73
Height [°]	18.0	18.0	19.0	20.0	20.0	27.0	28.0	28.0	29.0	32.0	34.0	38.0	39.0	39.0
Azimuth [°]	74	75	76	78	79	80	101	102	103	105	106	107	109	127
Height [°]	40.0	40.0	41.0	42.0	42.0	43.0	42.0	42.0	41.0	40.0	40.0	39.0	38.0	38.0
Azimuth [°]	129	135	143	144	150	151	154	155	156	169	170	177	178	179
Height [°]	36.0	31.0	31.0	30.0	24.0	24.0	22.0	20.0	20.0	15.0	15.0	11.0	12.0	12.0

Sun Paths (Height / Azimuth diagram)

Fixed plane, Tilts/azimuths: 32°/ 0°



Main results

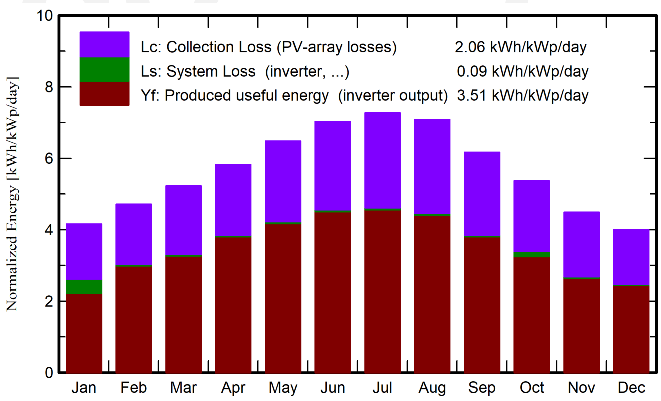
System Production

Produced Energy	19205923 kWh/year	Specific production	1281 kWh/kWp/year
Used Energy	26989774 kWh/year	Perf. Ratio PR	62.05 %
		Solar Fraction SF	32.65 %

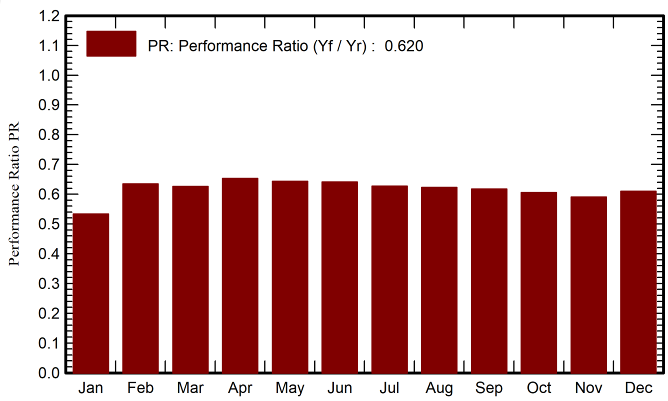
Economic evaluation

Investment		Yearly cost		LCOE	
Global	18,075,698.11 GBP	Annuities	763,297.13 GBP/yr	Energy cost	0.06 GBP/kWh
Specific	1.21 GBP/Wp	Run. costs	191,538.26 GBP/yr		
		Payback period	20.4 years		

Normalized productions (per installed kWp)



Performance Ratio PR



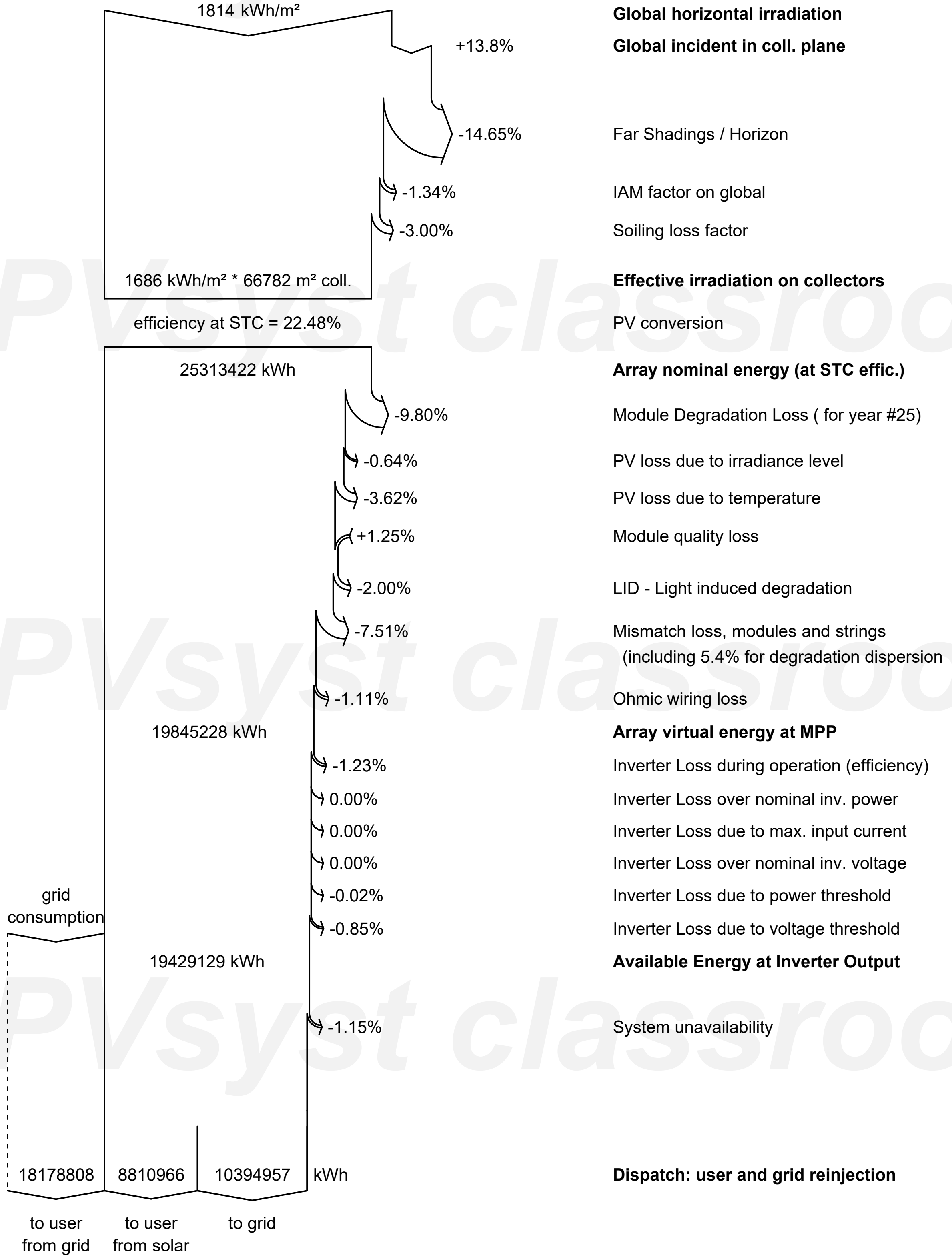
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	80.3	29.61	12.01	129.0	101.0	1220475	3010997	639351	391884	2371646
February	95.1	39.97	12.67	132.1	105.7	1273309	2747802	762817	493700	1984985
March	136.2	64.48	14.85	161.9	128.9	1540111	2913004	916487	604253	1996517
April	165.6	75.00	16.55	174.7	146.3	1734277	2721001	988180	723910	1732821
May	209.2	76.44	19.43	201.0	168.1	1964431	2026989	816185	1123486	1210805
June	230.2	71.13	22.17	210.8	178.5	2053424	1745004	720789	1306727	1024214
July	241.2	63.97	24.61	225.3	190.3	2144356	1924996	772604	1345765	1152392
August	212.9	66.39	24.86	219.5	184.3	2075762	2060987	807228	1243510	1253759
September	159.1	55.46	22.08	185.0	151.4	1734653	1971003	703740	1008739	1267263
October	123.6	46.04	19.41	166.4	134.8	1579496	2159004	658571	851751	1500433
November	86.9	30.89	15.19	134.7	101.6	1208499	2060997	567105	624882	1493892
December	73.6	28.04	13.13	124.1	95.2	1149456	1647990	457909	676350	1190080
Year	1813.9	647.41	18.11	2064.4	1686.1	19678247	26989774	8810966	10394957	18178808

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

Loss diagram



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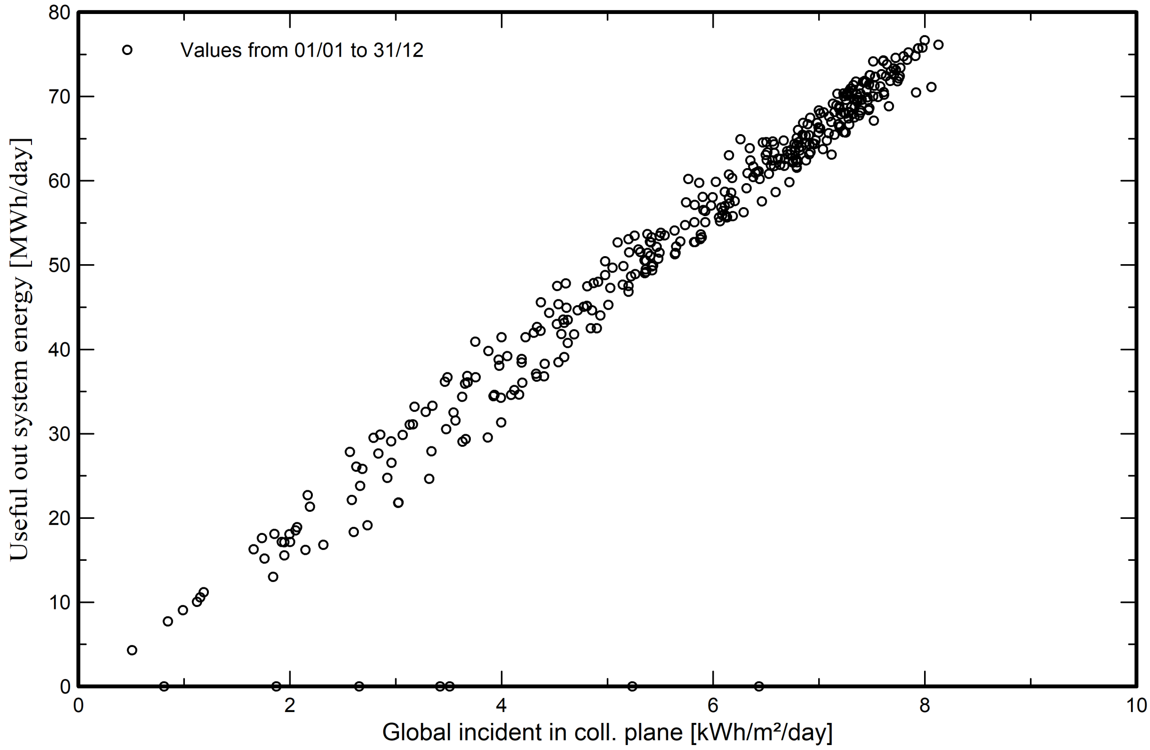
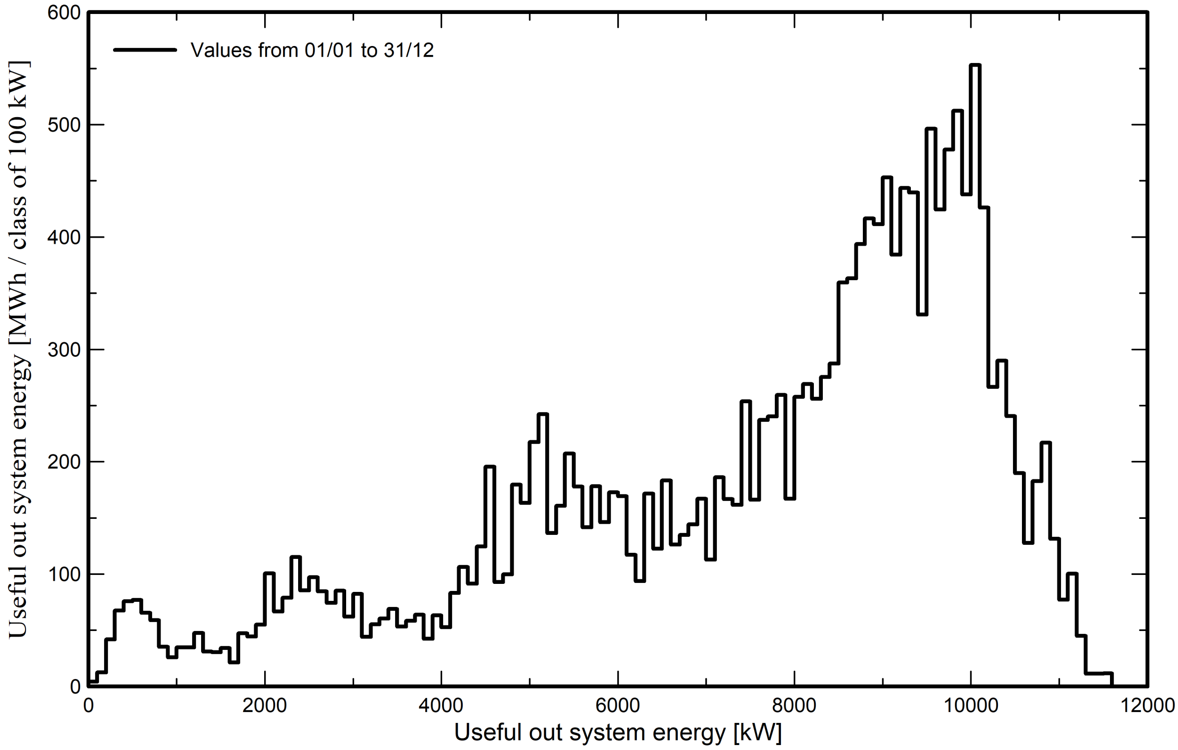
Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



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University of Edinburgh (United kingdom)

Aging Tool

Aging Parameters

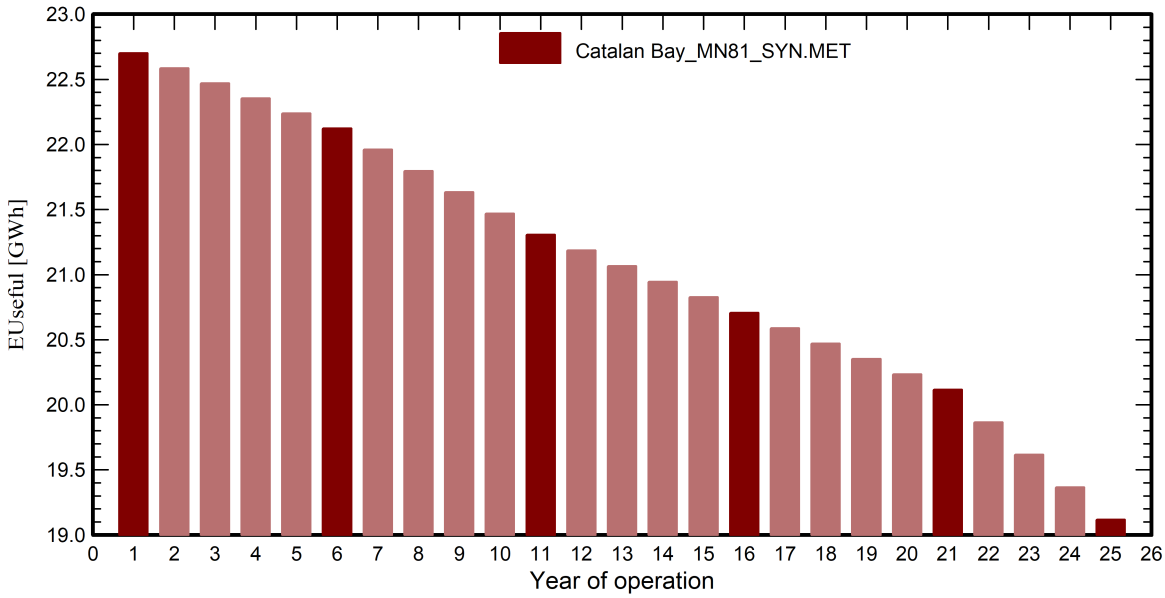
Time span of simulation 25 years

Module average degradation
Loss factor 0.4 %/year

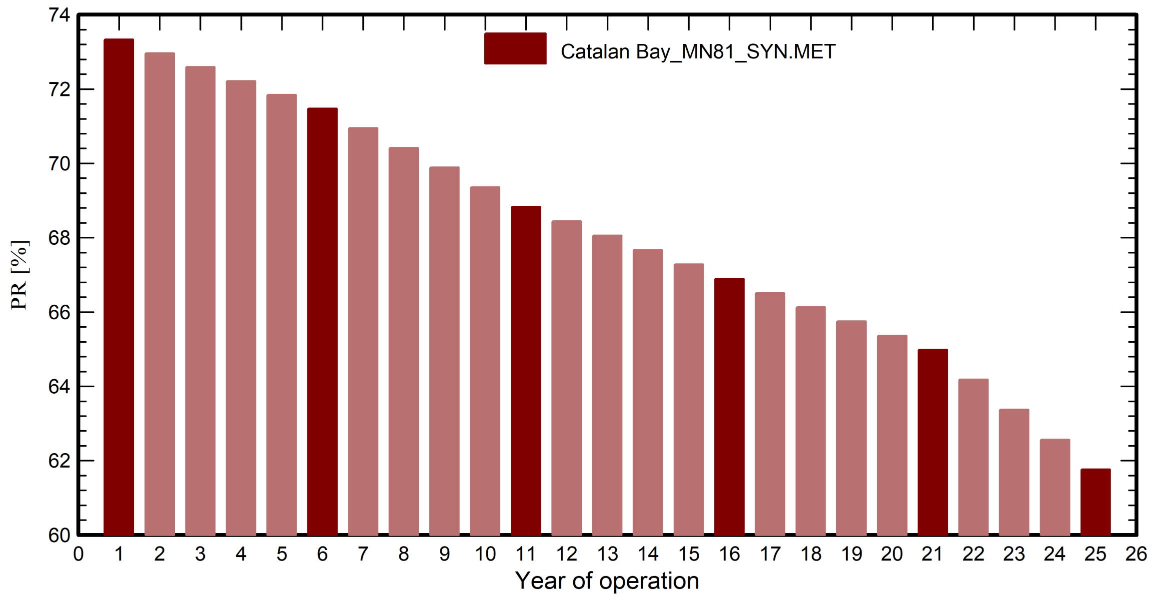
Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation
Catalan Bay MN81 SYN
Years reference year

Useful out system energy



Performance Ratio



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Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation

Catalan Bay MN81 SYN

Years reference year

	EUseful	PR	PR loss
Year	GWh	%	%
1	22.70	73.33	-0.25
2	22.58	72.96	-0.76
3	22.47	72.59	-1.27
4	22.35	72.22	-1.77
5	22.24	71.84	-2.28
6	22.12	71.47	-2.78
7	21.96	70.94	-3.50
8	21.80	70.41	-4.22
9	21.63	69.89	-4.94
10	21.47	69.36	-5.66
11	21.31	68.83	-6.38
12	21.19	68.44	-6.91
13	21.07	68.05	-7.43
14	20.95	67.67	-7.96
15	20.83	67.28	-8.49
16	20.71	66.89	-9.01
17	20.59	66.51	-9.53
18	20.47	66.13	-10.05
19	20.35	65.75	-10.57
20	20.23	65.36	-11.09
21	20.11	64.98	-11.61
22	19.87	64.18	-12.71
23	19.62	63.37	-13.80
24	19.37	62.57	-14.90
25	19.12	61.76	-15.99

Single-line diagram

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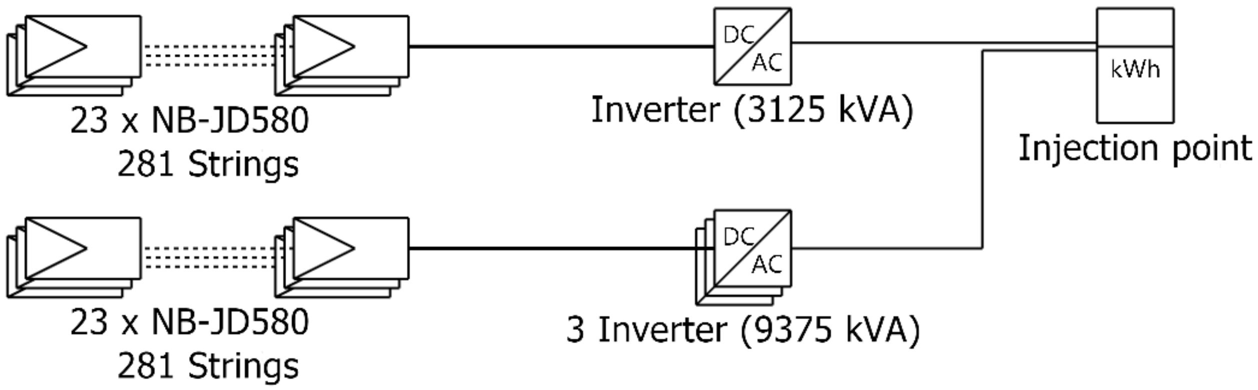


PV module	NB-JD580
Inverter	SG3125-HV-20
String	23 x NB-JD580

Gibraltar_Fixed_Solar_Panel	University of Edinburgh (United kingd
VC6 : Gibraltar Report Simulation - Simulation for year no 25	28/03/24

Cost of the system

Installation costs

Item	Quantity units	Cost GBP	Total GBP
PV modules			
NB-JD580	25852	172.00	4,446,544.00
Inverters			
SG3125-HV-20	4	20,000.00	80,000.00
Other components			
Wiring	15	58,491.46	877,371.90
Monitoring system, display screen	15	7,311.43	109,671.49
Measurement system, pyranometer	15	10,967.15	164,507.23
Studies and analysis			
Engineering	1	40,000.00	40,000.00
Installation			
Transport	1	100,000.00	100,000.00
Settings	15	694,586.09	10,418,791.37
Grid connection	15	54,835.74	822,536.16
Insurance			
Building insurance	1	50,000.00	50,000.00
Transport insurance	1	50,000.00	50,000.00
Land costs			
Land preparation	15	731.14	10,967.15
Taxes			
VAT	1	0.00	905,308.80
		Total	18,075,698.11
		Depreciable asset	4,526,544.00

Operating costs

Item	Total GBP/year
Maintenance	
Provision for inverter replacement	4,000.00
Salaries	73,479.90
Repairs	15,354.01
Land rent	32,901.45
Insurance	
Facilities insurance	21,934.30
Administrative, accounting	43,868.60
Total (OPEX)	191,538.26

System summary

Total installation cost	18,075,698.11 GBP
Operating costs	191,538.26 GBP/year
Useful energy from solar	8811 MWh/year
Energy sold to the grid	10395 MWh/year
Cost of produced energy (LCOE)	0.0628 GBP/kWh

Financial analysis

Simulation period

Project lifetime	25 years	Start year	2025

Income variation over time

Inflation	0.00 %/year
Production variation (aging)	Aging tool results
Discount rate	0.00 %/year

Income dependent expenses

Income tax rate	20.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (GBP)	Depreciable (GBP)
PV modules				
NB-JD580	Straight-line	20	0.00	4,446,544.00
Inverters				
SG3125-HV-20	Straight-line	20	0.00	80,000.00
		Total	0.00	4,526,544.00

Financing

Own funds	4,075,698.11 GBP	
Subsidies	1,000,000.00 GBP	
Loan - Redeemable with fixed annuity - 25 years	13,000,000.00 GBP	Interest rate: 3.20%/year

Electricity sale

Feed-in tariff	0.15000 GBP/kWh
Duration of tariff warranty	20 years
Annual connection tax	0.00 GBP/kWh
Annual tariff variation	0.0 %/year
Feed-in tariff decrease after warranty	0.00 %

Self-consumption

Consumption tariff	0.00001 GBP/kWh
Tariff evolution	0.0 %/year

Return on investment

Payback period	20.4 years
Net present value (NPV)	4,026,279.68 GBP
Internal rate of return (IRR)	7.59 %
Return on investment (ROI)	23.6 %

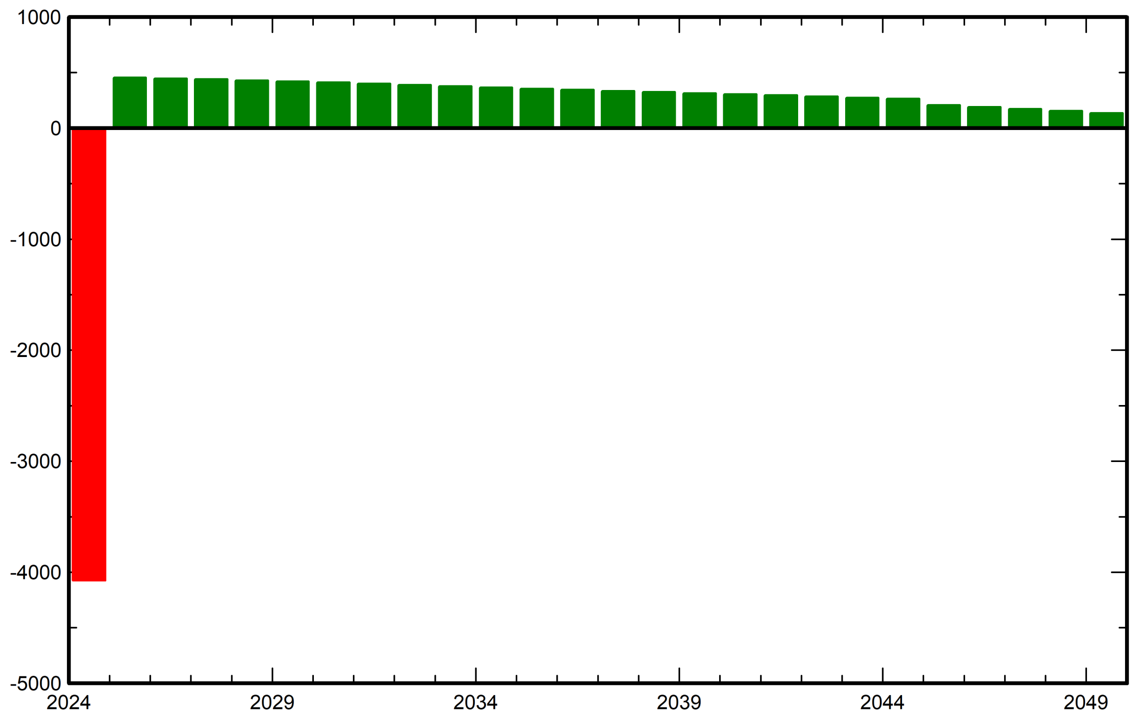
Financial analysis

Detailed economic results (kGBP)

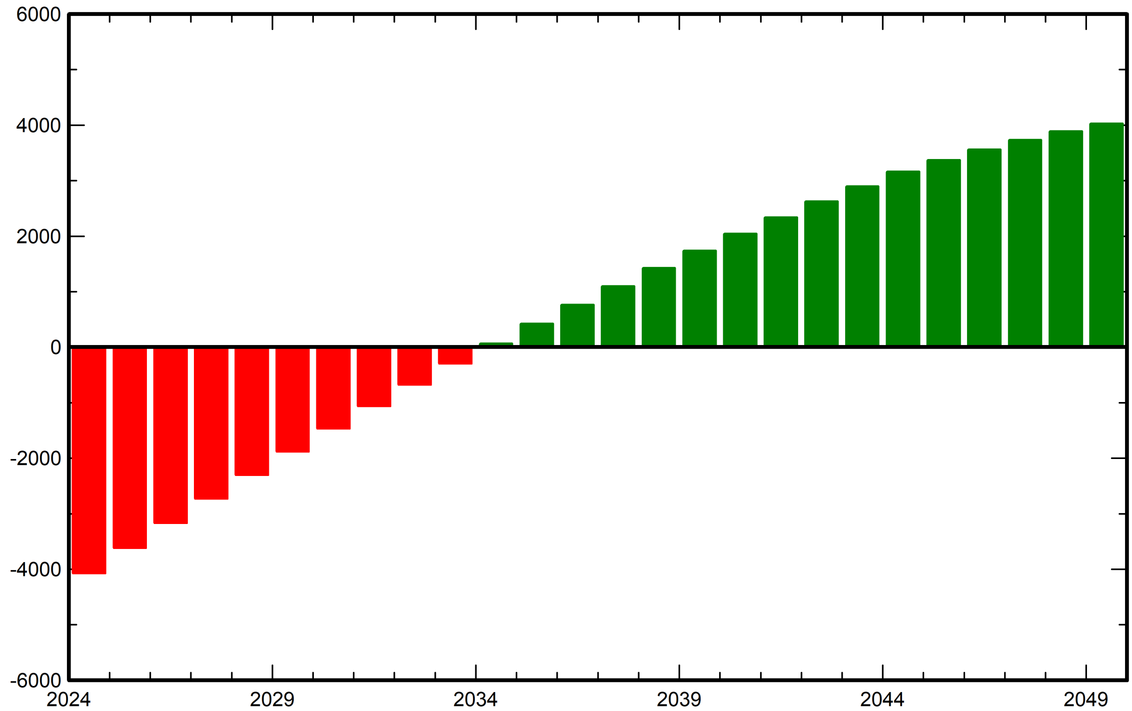
Year	Electricity sale	Own funds	Loan principal	Loan interest	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Self-cons. saving	Cumul. profit	% amorti.
0	0	4,075,698	0	0	0	0	0	0	0	0	-4,075,698	0.0%
1	1,555,302	0	347,297	416,000	191,538	226,327	721,437	144,287	456,179	88	-3,619,431	4.7%
2	1,547,409	0	358,411	404,886	191,538	226,327	724,657	144,931	447,642	87	-3,171,701	9.4%
3	1,539,516	0	369,880	393,417	191,538	226,327	728,233	145,647	439,034	87	-2,732,580	14.2%
4	1,531,623	0	381,716	381,581	191,538	226,327	732,176	146,435	430,352	87	-2,302,141	18.9%
5	1,523,730	0	393,931	369,366	191,538	226,327	736,498	147,300	421,595	86	-1,880,460	23.7%
6	1,515,837	0	406,537	356,761	191,538	226,327	741,211	148,242	412,760	86	-1,467,615	28.5%
7	1,504,624	0	419,546	343,751	191,538	226,327	743,007	148,601	401,187	85	-1,066,343	33.3%
8	1,493,411	0	432,971	330,326	191,538	226,327	745,219	149,044	389,531	84	-676,727	38.1%
9	1,482,197	0	446,826	316,471	191,538	226,327	747,861	149,572	377,790	84	-298,853	42.9%
10	1,470,984	0	461,125	302,172	191,538	226,327	750,946	150,189	365,960	83	67,189	47.8%
11	1,459,771	0	475,881	287,416	191,538	226,327	754,489	150,898	354,038	82	421,309	52.7%
12	1,451,556	0	491,109	272,188	191,538	226,327	761,503	152,301	344,420	82	765,812	57.5%
13	1,443,342	0	506,824	256,473	191,538	226,327	769,003	153,801	334,705	82	1,100,599	62.5%
14	1,435,127	0	523,043	240,254	191,538	226,327	777,007	155,401	324,890	81	1,425,570	67.4%
15	1,426,912	0	539,780	223,517	191,538	226,327	785,530	157,106	314,971	81	1,740,621	72.4%
16	1,418,697	0	557,053	206,244	191,538	226,327	794,588	158,918	304,944	80	2,045,646	77.5%
17	1,410,603	0	574,879	188,418	191,538	226,327	804,319	160,864	294,904	80	2,340,629	82.6%
18	1,402,508	0	593,275	170,022	191,538	226,327	814,621	162,924	284,749	79	2,625,457	87.7%
19	1,394,414	0	612,260	151,037	191,538	226,327	825,511	165,102	274,476	79	2,900,012	92.9%
20	1,386,319	0	631,852	131,445	191,538	226,327	837,008	167,402	264,082	78	3,164,172	98.2%
21	1,378,224	0	652,071	111,226	191,538	0	1,075,460	215,092	208,297	78	3,372,547	103.2%
22	1,361,141	0	672,938	90,359	191,538	0	1,079,243	215,849	190,457	77	3,563,081	108.3%
23	1,344,057	0	694,472	68,825	191,538	0	1,083,694	216,739	172,483	76	3,735,640	113.3%
24	1,326,974	0	716,695	46,602	191,538	0	1,088,833	217,767	154,372	75	3,890,087	118.4%
25	1,309,891	0	739,629	23,668	191,538	0	1,094,684	218,937	136,118	74	4,026,280	123.6%
Total	**36,114,170**	**4,075,698**	**13,000,000**	**6,082,428**	**4,788,456**	**4,526,544**	**20,716,741**	**4,143,348**	**8,099,937**	**2,041**	**4,026,280**	**123.6%**

Financial analysis

Yearly net profit (kGBP)



Cumulative cashflow (kGBP)



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CO_2 Emission Balance

Total: 298711.7 tCO_2

Generated emissions

Total: 80235.32 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 436743.3 tCO_2

System production: 19205.95 MWh/yr

Grid Lifecycle Emissions: 758 gCO_2/kWh

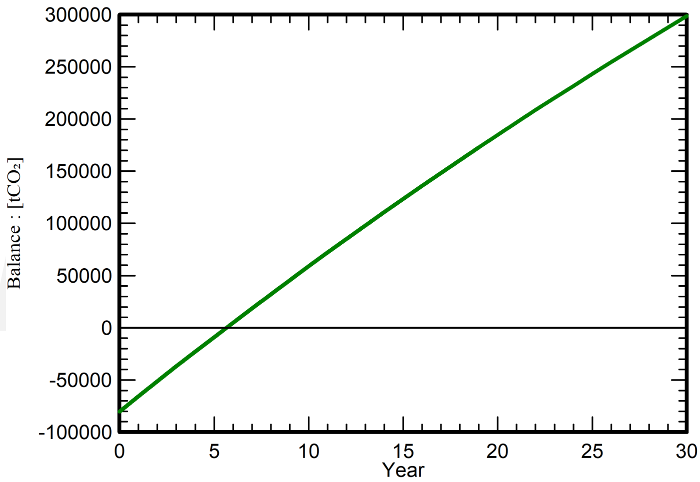
Source: IEA List

Country: Gibraltar

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	2126 kgCO2/kWp	36250 kWp	77072400
Supports	5.06 kgCO2/kg	625000 kg	3159913
Inverters	501 kgCO2/units	6.00 units	3005